

Answer **all** the questions.

1. (a) The table shows the resident population of Singapore in three years.

Year	2000	2010	2020
Resident population	3 273 363	3 771 721	4 044 210

- (i) Write the 2010 resident population in standard form, correct to three significant figures.

Answer [1]

- (ii) Calculate the percentage increase in the resident population from 2000 to 2020.

Answer % [2]

- (iii) In 2000, 18.7% of the total population of Singapore was non-resident.
Calculate the total population of Singapore in 2000.

Answer [2]

- (b) Population density is defined as the number of people per square kilometre.
In 2020, the total population density of Singapore was 7810 people per square kilometre.
In 2020, the total population of Singapore was 5.69 million.
Calculate the area of Singapore.

Answer km² [2]

- (c) The diameter of a red blood cell is 7.8×10^{-6} m.
 In a scale drawing, the diameter of a red blood cell is 3.9 cm.
 (i) Find the scale used for the drawing.
 Give your answer in the form $n : 1$.

Answer : 1 [2]

- (ii) Drawn to the same scale as the red blood cell, the diameter of a blood platelet is 8 mm.
 Find the actual diameter of the blood platelet.
 Give your answer in millimetres in standard form.

Answer mm [1]

2. (a) Solve $6 - 7x = 5(1 - 2x)$.

Answer $x =$ [2]

- (b) Solve the inequality $5 - y > 7$.

Answer [1]

(c) $c = \frac{1}{b} + \frac{a}{3 - a}$

- (i) Find c when $a = 2$ and $b = 4$.

Answer $c =$ [1]

- (ii) Rearrange the formula to make a the subject.

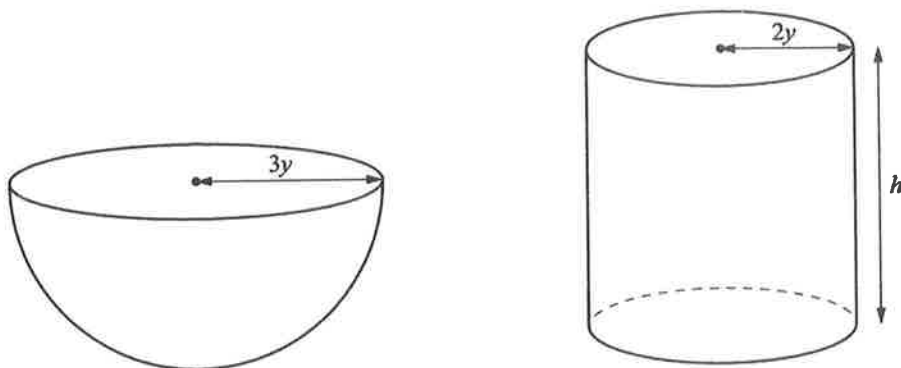
Answer $a = \dots\dots\dots$ [3]

- (d) Solve the equation $\frac{x}{2x-1} - \frac{6}{4-x} = 3$.

Give your solutions correct to two decimal places.

Answer $x = \dots\dots\dots$ or $\dots\dots\dots$ [5]

3.



The diagram shows a solid hemisphere and a solid cylinder.

The hemisphere has radius $3y$ cm.

The cylinder has radius $2y$ cm and height h cm.

(a) Show that the total surface area of the hemisphere is $27\pi y^2$ cm².

Answer

[2]

(b) The total surface area of the cylinder is equal to the total surface area of the hemisphere.
Find h in terms of y .

Answer $h = \dots\dots\dots$ [3]

- (c) The volume of the hemisphere is 500 cm^3 .
Calculate the volume of the cylinder.

Answer cm^3 [4]

4. (a) Complete the table of values for $y = \frac{2}{x^2} + 3x - 1$.
Values are given to one decimal place where appropriate.

x	-3	-2	-1	-0.5	0.5	1	1.5	2	3
y		-6.5	-2	5.5	8.5	4	4.4	5.5	8.2

[1]

- (b) On the grid, draw the graph of $y = \frac{2}{x^2} + 3x - 1$ for $-3 \leq x \leq 3$.

[3]

- (c) (i) On the same grid, draw the graph of $2y - 5x = 7$ for $-3 \leq x \leq 3$.

[2]

- (ii) Write down the x -coordinates of the points where the line intersects the curve.

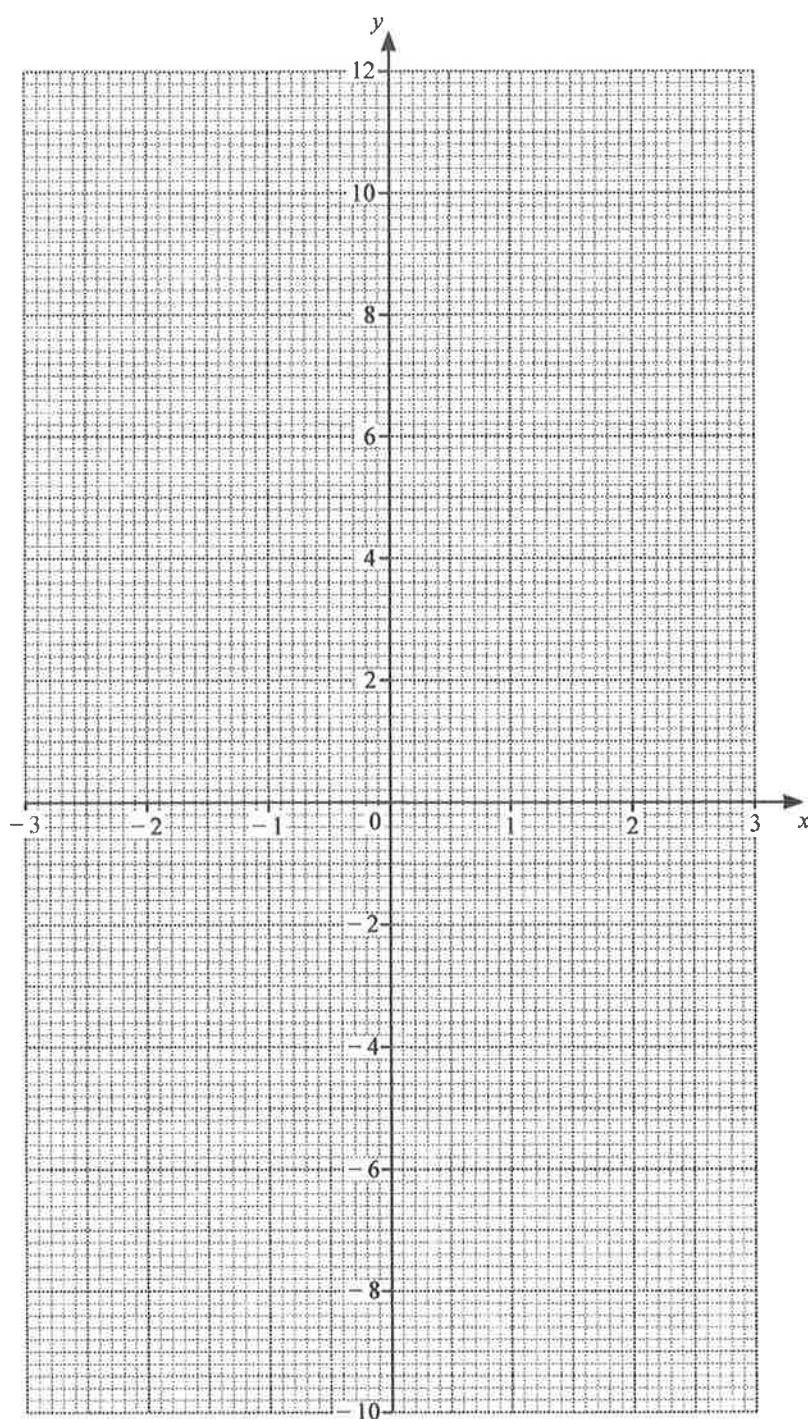
Answer $x = \dots\dots\dots$ and $\dots\dots\dots$ [2]

(iii) These values of x are solutions of the equation $x^3 + Ax^2 + B = 0$.

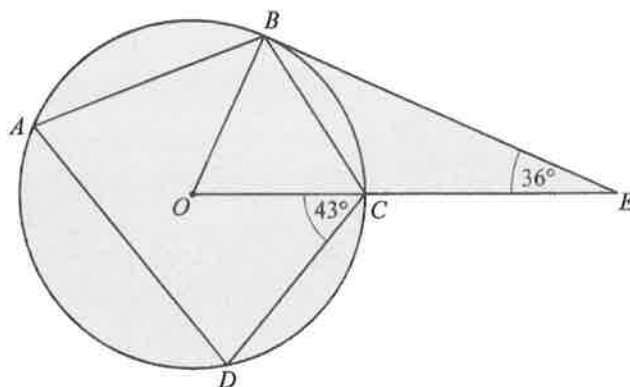
Find the value of A and the value of B .

Answer $A = \dots\dots\dots$

$B = \dots\dots\dots$ [3]



5.



A, B, C and D are points on the circle, centre O .

EB is a tangent to the circle at B and OE is a straight line that passes through point C .

Angle $OEB = 36^\circ$ and angle $OCD = 43^\circ$.

(a) Find angle BAD .

Give reasons for each step of your working.

Answer Angle $BAD = \dots\dots\dots [4]$

(b) The radius of the circle is 8 cm.

Calculate the perimeter of the shaded shape.

Answer $\dots\dots\dots$ cm [4]

6. (a) $\mathbf{p} = \begin{pmatrix} -4 \\ 5 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ -3 \end{pmatrix}$.

X is the point $(7, 3)$ and Y is the point $(3, 4)$.

$\overrightarrow{XY} = m\mathbf{p} + n\mathbf{q}$, where m and n are constants.

Find the value of m and the value of n .

Answer $m = \dots\dots\dots$

$n = \dots\dots\dots$ [4]

(b) A is the point $(4, 6)$, B is the point $(-5, -3)$ and C is the point $(8, -4)$.

D is the point $(-1, y)$ which lies on the line AB .

(i) Find the value of y .

Answer $y = \dots\dots\dots$ [1]

(ii) Find the length of the line DC .

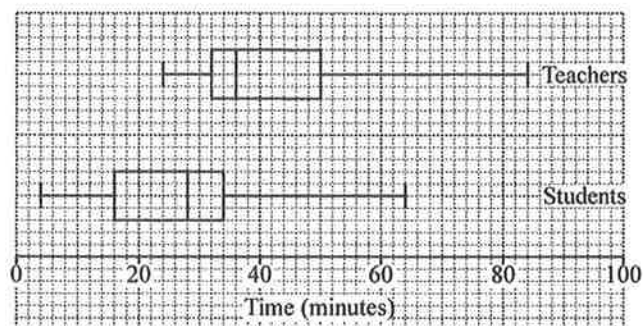
Answer $\dots\dots\dots$ [2]

- (iii) C is the point on the line AE such that $AC : AE = 2 : 3$.

Find the equation of the line BE .

Answer [4]

7. (a) The students and teachers at a school record the time it took them to travel to school.
The box-and-whisker plots show the distributions of the results.



- (i) There are 18 teachers who took more than 50 minutes to travel to school.
Work out the number of teachers in the school.

Answer [1]

- (ii) Make a comment comparing the averages and a comment comparing the distributions of the times taken by the students and the teachers.

Use figures to support your answers.

1.
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.....
2.
.....
.....

[3]

- (b) In a class of 24 students, 8 walk to school, 6 cycle to school and the rest travel to school by bus.

- (i) One student is selected from the class at random.

Find the probability that the student travels to school by bus.

Answer [1]

- (ii) Two students are selected from the class at random.

Find the probability that they both use the same type of transport to travel to school.

Answer [2]

(iii) Three of the students are selected from the class at random.

Find the probability that two of them walk to school and the other does not.

Answer [2]

8. (a) The diagram shows part of a number grid.

1	2	3	4	5	6	7	8
9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24
25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48

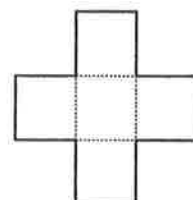
A cross outlining five numbers, as shown, is placed on the grid.

(i) The cross is placed in a different position on the grid.

The sum of the largest and smallest of the numbers outlined is 204.

Complete the cross to show the numbers outlined.

Answer



[3]

- (ii) The cross can be placed anywhere on the grid.

The difference between the product of the left and right numbers and the product of the top and bottom numbers is found.

Show that this difference is always 63.

Answer

[3]

- (b) The n th term of a sequence is given by $T_n = \frac{n^2 - 1}{3n - 1}$.

The term $T_k = \frac{15}{4}$.

Find the value of T_{k+1} .

Answer $T_{k+1} = \dots\dots\dots$ [4]

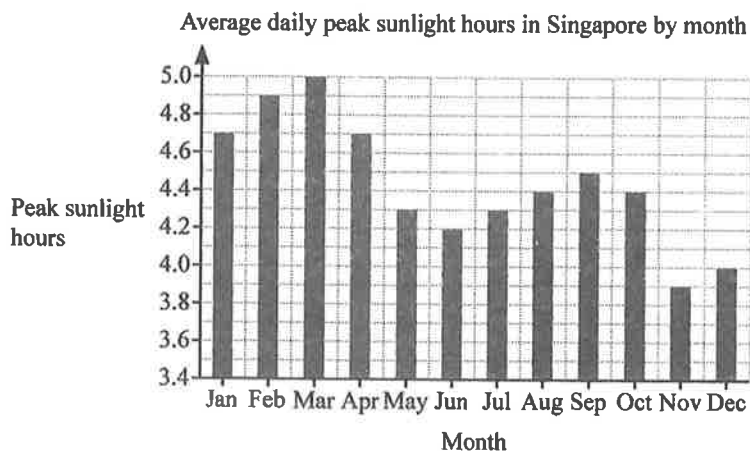
9. Chen is considering installing solar panels at his home.

He wants to estimate the area of solar panels he will need to generate the amount of electricity used by his household.

He finds this information on the internet.

- Find the average amount of electricity, measured in kilowatt hours (kWh), used in one day
- Find the number of peak sunlight hours you receive per day
- Solar panel output in daily watt hours = solar panel watts $\times$ peak sunlight hours $\times$ 75%
- All solar panels should be of the same type
- 1 kWh is the amount of energy used in 1 hour by an appliance with a power rating of 1 kW
- 1 kW = 1000 W

He does some more research about peak sunlight hours and the specifications of common solar panels.



Type	Solar panel wattage	Length of panel	Width of panel	Mass of panel
A	300 W	1.46 m	1.05 m	18 kg
B	370 W	1.96 m	1.00 m	23 kg
C	390 W	1.98 m	1.00 m	28 kg
D	400 W	1.69 m	1.05 m	19 kg

- (a) A games console has a power rating of 165 W.

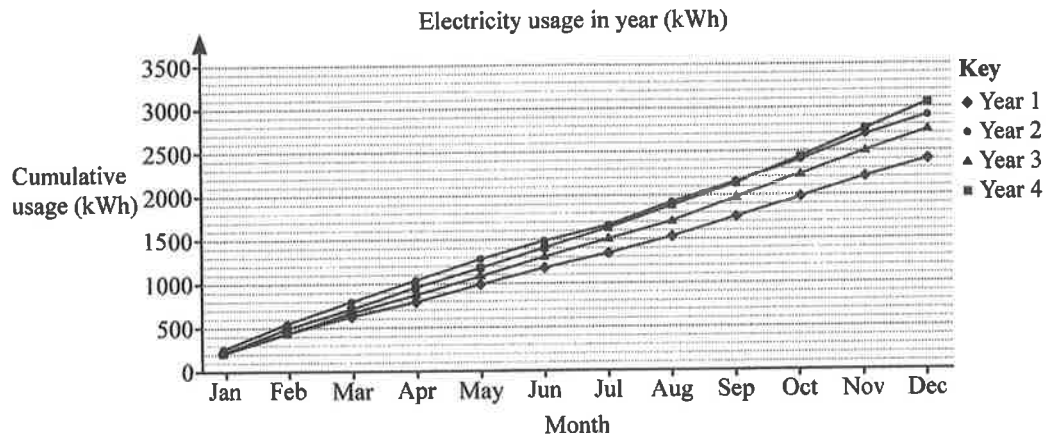
Calculate the amount of energy, in kWh, used in 5 hours by the games console.

Answer kWh [1]

- (b) Assuming all months have an equal number of days, calculate the average daily peak sunlight hours in Singapore over the 12-month period.

Answer hours [2]

- (c) This graph shows the electricity used by Chen's household for the past four years.



Find an estimate of the total area of solar panels Chen will need for his home.

Justify any decisions you make and show your calculations clearly.

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..... [7]